

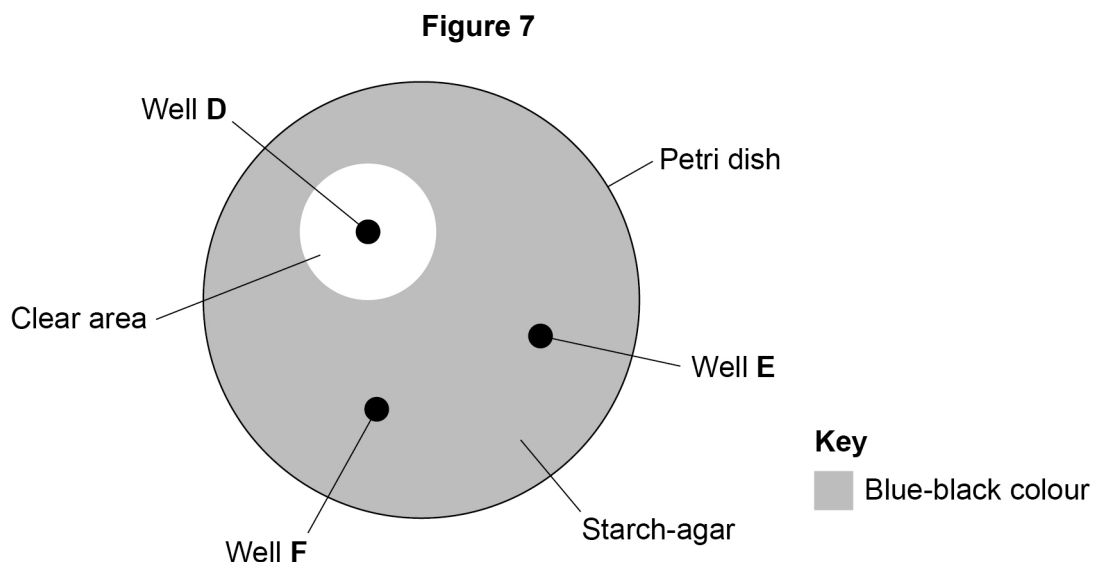
0 5

A student investigated the activity of the enzyme amylase. He cut three identical wells (**D**, **E** and **F**) in starch-agar in a Petri dish. He added 0.2 cm^3 of:

- amylase solution to well **D**
- boiled amylase solution to well **E**
- water to well **F**.

After 60 minutes, he covered the starch-agar with iodine solution.

Figure 7 shows his results.

**0 5 . 1**

Explain the appearance of the agar in the clear area surrounding well **D**.

[2 marks]



0 5 . 2

What can you conclude about the activity of amylase from the appearance of the agar surrounding well **E** and well **F** in **Figure 7**?

[2 marks]

0 5 . 3

The student cut out a piece of agar from the clear area surrounding well **D**. He obtained a solution of the substances contained in this piece of agar.

Describe a different biochemical test the student could use with this solution to confirm that amylase had affected the starch in the clear area surrounding well **D**.

[2 marks]

Question 5 continues on the next page

Turn over ►

The diameter of the clear area around well **D** is 18 mm

In a different investigation, the student prepared a dilution of the amylase solution. He did this by mixing amylase solution and water in the volumes shown in **Table 1**.

Table 1

Amylase solution / cm ³	Water / cm ³
1.6	2.4

He prepared a starch-agar Petri dish identical to **Figure 7**, but with a single well. He added 0.2 cm³ of the diluted amylase solution to this well and left the Petri dish for 60 minutes.

0 5 . 4

Use all of this information to predict the diameter of the clear area that will form around the well containing the diluted amylase solution.

Give your answer to the nearest whole number.

Show your working.

[2 marks]

Answer _____ mm

0 5 . 5

The student used a ruler to measure the diameter in mm of the clear area around well **D** in **Figure 7**.

Use this information to explain why the answer to Question **05.4** should be given to the nearest whole number.

[1 mark]

